

TY E&TC Electronic system design (PE)

Course code: UETPE0513

Unit 1

1. What is reliability? Explain exponential law of reliability? Which factor affects reliability of the equipment?
2. Discuss the causes of failures during the Infant Mortality, Useful Life, and Wear-out periods of the Bathtub Curve.
3. Explain how System Performance Matrix and Design Matrix is used for decision-making during product development and system design.
4. An inverting amplifier with $R_1=10K$ and $R_f=20K$ and $V_i=3V$, drives a $2k$ load a. Assume $I_Q=0.5ma$, find i_{cc} , i_{ee} and i_o b. Find the power dissipated inside the Op-Amp
5. Explain the need for TTL-CMOS interfacing. Discuss TTL-to-CMOS and CMOS-to-TTL interfacing techniques with suitable circuits.

Unit 2

1. Design a V to I converter that delivers $4ma$ to $20ma$ current to a floating load, over its input voltage range of 0 to $4Volts$
2. Design an on-off controller to control air pressure in a pneumatic tank. The pressure sensor delivers linear output between $0-2Volts$ DC over pressure range of 0 to $10kg/cm^2$. The pressure in the tank is to be controlled at $6kg/cm^2$ with hysteresis of $\pm 0.2kg/cm^2$. The On-Off controller output should be 0 to $12 VDC$ for On and off respectively. Draw a neat circuit diagram and design the on-off controller.
3. Design a signal conditioner circuit for K type Thermocouple to get output sensitivity of $10mv/oC$ and give $0mv$ at $0oC$. Make provision of cold Junction compensation.
4. Derive the expression for the gain of an Instrumentation Amplifier and discuss its characteristics and applications.
5. Discuss important specifications of the load cell.

Unit 3

1. Design static display for a product counter
2. What are the objectives of dynamic 7 segment display design? How it is achieved? Design microcontroller based dispensed petrol / diesel volume measurement system with resolution of $10ml$ and maximum dispensed volume to be $999.99Ltrs$.
3. Discuss important specifications of ADC/DAC.
4. A 10 bit successive approximation ADC is supplied with 4.094 volts reference voltage. Find resolution of ADC. Design a noninverting amplifier with gain 100 to amplify sensor output. Make sure that errors developed in the noninverting amplifier due to input offset voltage, input bias current and input offset current of the selected Opamp should not generate errors in the measurement due to the effect of the Opamps parameter?
5. Explain the construction and working principle of a Capacitive Touch Screen Interface.

The last date to submit the assignment is 12th Sept for Unit 1 and 2

The last date to submit the assignment is 19th Sept for Unit 3

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